UNDERSTANDING SOCIAL MOVEMENT MESSAGING FROM A COLLECTION OF TOPICAL IMAGES: USING MULTIMODAL SENTIMENT ANALYSIS FOR NARRATIVES IN STRUCTURED GRAPHS

by

Laura Werthmann Dozal

Copyright © 2026

A Dissertation Submitted to the Faculty of the

COLLEGE OF INFORMATION SCIENCE

In Partial Fulfillment of the Requirements

For the Degree of

DOCTOR OF PHILOSOPHY

INFORMATION SCIENCE

In the Graduate College

THE UNIVERSITY OF ARIZONA

2026

PREVIEW

THE UNIVERSITY OF ARIZONA
GRADUATE COLLEGE

As members of the Dissertation Committee, we certify that we have read the dissertation prepared by: Laura Werthmann Dozal

titled: UNDERSTANDING SOCIAL MOVEMENT MESSAGING FROM A COLLECTION OF TOPICAL IMAGES: USING MULTIMODAL SENTIMENT ANALYSIS FOR NARRATIVES IN STRUCTURED GRAPHS

and recommend that it be accepted as fulfilling the dissertation requirement for the Degree of Doctor of Philosophy.

Jamie Lee — Date: Mar 10, 2026

Zack Lischer-Katz (Mar 16, 2026 12:40:01 PDT)
Zack Lischer-Katz — Date: Mar 16, 2026

Charles Gomez
Charles Gomez — Date: Mar 11, 2026

Clayton Morrison — Date: Mar 10, 2026

Final approval and acceptance of this dissertation is contingent upon the candidate's submission of the final copies of the dissertation to the Graduate College.

I hereby certify that I have read this dissertation prepared under my direction and recommend that it be accepted as fulfilling the dissertation requirement.

Jamie Lee — Date: Mar 10, 2026
College of Information Science

PREVIEW

ACKNOWLEDGEMENTS

This has been an invaluable investment in myself and my future, I am deeply grateful to everyone who supported me along the way.

To my advisor, Dr. Jaime Lee, for your guidance and for reminding me to slow down and think deeply. To my committee members, Zack Lischer-Katz, Clayton Morrison, and Charlie Gomez, for your thoughtful feedback and recommendations during the various stages of my dissertation.

Thank you to the University of Arizona Libraries data team where Tuesday morning R Labs helped me learn how to code with great patience and overall support. My academic computational experience would not have been fruitful without all your expertise and resources.

Thank you to the Jetstream2 and CyVerse team who supported me at the end of my dissertation journey. You made sure to support me at the drop of a hat. Much of the support and GPU resources were provided by Jetstream2 at Indiana University through allocation [CIS250051] from the Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) program, which is supported by National Science Foundation grants #2138259, #2138286, #2138307, #2137603, and #2138296.

Thank you to my family who were patient and stuck around as I endlessly worked through many holidays and special events.

Thank you to my best friend and support system Maleny Martinez and her family, you and Eva's love and friendship are invaluable.

PREVIEW

Thank you to my Texas friends and family Mabel, Sandi, Karla, Victoria, Dan, Ana, Katrina, Fallon, Luisana, Oscar, Elmer, Grecia, Alex, and Yazmin.

Thank you to my CCT Family, Celina, Areej, Mariam, Lisa-Marie, Ramon, Elerie, Linda, Megan, Mike, Sam, Chris, Bryan, Will, Molly, Martin, and Siying. I truly could not have done this without your inspiration.

Thank you to my PhD Peers who were my life support during the who experience, Sarah, Liliana, Ana, Farzaneh, Yenni, Salena, Gaby, Mac, Kelsey, and Loren + family.

PREVIEW

TABLE OF CONTENTS

PREVIEW

PREVIEW

PREVIEW

PREVIEW

PREVIEW

LIST OF FIGURES

PREVIEW

PREVIEW

PREVIEW

PREVIEW

PREVIEW

PREVIEW

PREVIEW

PREVIEW

LIST OF TABLES

PREVIEW

PREVIEW

PREVIEW

ABSTRACT

This dissertation research explores how fragmented social media messaging about a particular social movement (the anti-feminicide movement) is represented as an overall narrative. Social movements have been using social media to share their goals, values, and grievances with the world for the past decade, which spread throughout platforms usually in visual form, and across various perspectives. This messaging is sprinkled in with other news and cultural events that fail to provide a full narrative of what is actually happening. I analyze the visual and textual narrative representations of the online anti-feminicide movement in Mexico using narrative structure, network science, visual analysis, and a combination of multi-modal computer vision / natural language processing tools. The codebase and other resources can be found here: https://github.com/lwdozal/Narrative_Network_Pipeline

The overall research is a mixed-methods research project that answers the question, how can we identify a narrative structure from a collection of topical images using computational methods. It provides an in-depth review of the feminicide history in Mexico and its related anti-feminicide movement. It provides detailed descriptions of how the media has been framed since the 1990s using: Societal, Cultural, Social Movement, and Humanization. The analysis is completed in a three-step pipeline to build a narrative structure representative of the local semantics of image groups and an overall global summarization of the narrative structure as a whole.

This first step pulls feature embeddings from a collection of 16,657 Instagram images pertaining to the anti-feminicide movement in Mexico; and clusters them to identify common vector embedding groups within the data. Human-in-the-Loop (HITL) content analysis is applied to provide a qualitative understanding of the images within each cluster group and gain insight on the classification aspects of the images for downstream tasks. The next step takes these images and runs them through foundational vision transformer models, along with generative Multimodal LLLMs (M-LLMs). These models are tasked with generating two labels identified in the first set, and a description of each image. The outputs are compared and analyzed using multi-label classification metrics for machine learning models along with HITL discourse analysis of the categorization process. Step three uses the top labels created from the model with the best overall quantitative and HITL discourse evaluation metrics to build a network graph representing the various stages of semantics found within the image and its attributes 1) labels 2) descriptions and original post comments. The graph is analyzed to identify structural patterns of the network using traditional centrality measures and a modularity algorithm (Leiden).

The process identified three local features, Narrator, Action, and Subject and found that Narrator and Subject features were typically tied together as label attributes while the Action feature stood alone. Showing how the narrative of the social movement messages is structured by the generated labels. The graph is of labels, generated descriptions and original post comments

PREVIEW

analyzed to identify structural sentiment patterns of the network using traditional centrality measures and modularity models (Leiden algorithm) within and across the network. The overall graph structure is compared to benchmark datasets in a Graph Attention Network (GAT) model that uses a variational autoencoder to compare classifications of the collected annotated image data. In the results and discussion section of the dissertation, these quantitative findings are used in conjunction with a theoretical analysis of how the media frames the anti-feminicide movement in Mexico and implements an analysis review of the network's local and global semantics creating narrative structure for the collection of topical images.

PREVIEW

1 | Chapter - Introduction to Social Media News Fragmentation for Social Movements

As a global society living in the digital age of rapid information transfer, it's difficult to know what is actually going on around the world, or even in our backyard. We are social media content consumers who receive small bits of information that provide a story from one or multiple perspectives. The affordances of our preferred platforms dictate the information we get, but similar to news cycles, seemingly important information is spread rapidly and then pivots to a new more attention-grabbing story (Mourão, Brown, 2021). Unless a viewer does extra research, is honed in on the background, or understands the full context of a social media news event, it is difficult to understand the nuances and supporting implications of a viral news event.

This was my experience in August of 2019 as anti-feminicide protests crowded Mexico's capital city organized by grassroots activists who broadcasted their efforts on all social media platforms. These protests were followed by other collective action activities like federal building takeovers and glitter bombing, also broadcast online (Webb, 2023; Loyola-Hernández, 2025). Although the overall collective action for anti-feminicide was clear, it was hard to understand what was going on, why it started, and what people were thinking. The way my social media bubble represented these events, they mainly called for the same changes, expressed similar grievances, and framed personal experiences connected to the movement from different points of view. As someone born and raised on the U.S. Mexican border, particularly in the town bordering the first recorded femicide in Mexico, my interests in this topic piqued and my own personal research on collective action was done. But, as participants of an online society, some find it difficult to step off these platforms and investigate further. For most, the information provided is all a viewer receives and can be fragmented or split between other personal interest posts populating the feed. In this case, I argue that there are ways to get a grasp on the narrative of a social media news cycle, particularly the anti-feminicide social movement, by reviewing the narrative structure of its Instagram content and post metadata as a collection of topical images.

PREVIEW

1.1 Research Problem

Recent reports show that more Americans consume their news and information from social media (Leppert 2024). Particularly, 26% of adults under 30 years old regularly get their news from visual based content social media platform, Tik Tok (Leppert, 2024). This does not include other platforms like Instagram, Facebook, or Twitter where 50% of adults get their news from social media overall (Pew Research, 2024). As many of these platforms provide content referred from user feedback for general personalization, users are not provided with full information about an event (Pew Research, 2020). Most of the time the information comes from sources that mirror a user's preferences and activity on the platform (Mosseri, 2021). These sources are user generated content that have been shown to portray different behaviors than what might be observed in offline and non-media contexts. The Instagram search algorithms also apply content recommendations through metrics to provide similar suggestions for users similar to each other (Sanchez-Moreno et al, 2020; Mosseri, 2021). This information bubble can provide a biased perspective of a real-time event that could be impactful to a community (Chen et al, 2020) and in turn enable a fragmented interpretation of what is happening. Similarly, each platform brings about different information consumption tendencies and so each provides a unique perspective of an event (Pelletier et al, 2020). These perspectives might be conflicting or fragmented, and information of a current event could be distorted.

Conflicting and fragmented information is not ideal for social movement practitioners, especially when they use social media to spread their cause. Current communication practices of social movements almost always require an online presence to connect with members and potential supporters (Peterson-Salahuddin, C.,2022). As real-time visual content has become a main force for informational content on social media, social movements might not have full control over their messaging and in-turn, lose sight of the online narrative (Leppert, 2024; Earl, Garrett, 2016; Moore-Gilbert et al, 2017). The anti-feminicide movement in Mexico, for example, spreads across multiple online accounts and platforms, which initiate decentralized fragments of the movement's messaging goals (D'Ignazio, 2020). Much of the movement's content is a variation of informational, visual, call to action, or a combination, and can be

PREVIEW

Reproduced with permission of copyright owner. Further reproduction prohibited without permission.